HEAR FROM PEERS · CENTER FOR TEACHING EXCELLENCE · 28 OCTOBER 2026

# Building traceability into a course with AI

GEOG 476, GIS Practicum, Fall 2026

Daniel W. Goldberg · Department of Geography · daniel.goldberg@tamu.edu

# What traceability means in GEOG 476

## Outcomes

Every activity reaches a degree outcome

## Order

Nothing is due before it is taught

## Thinking

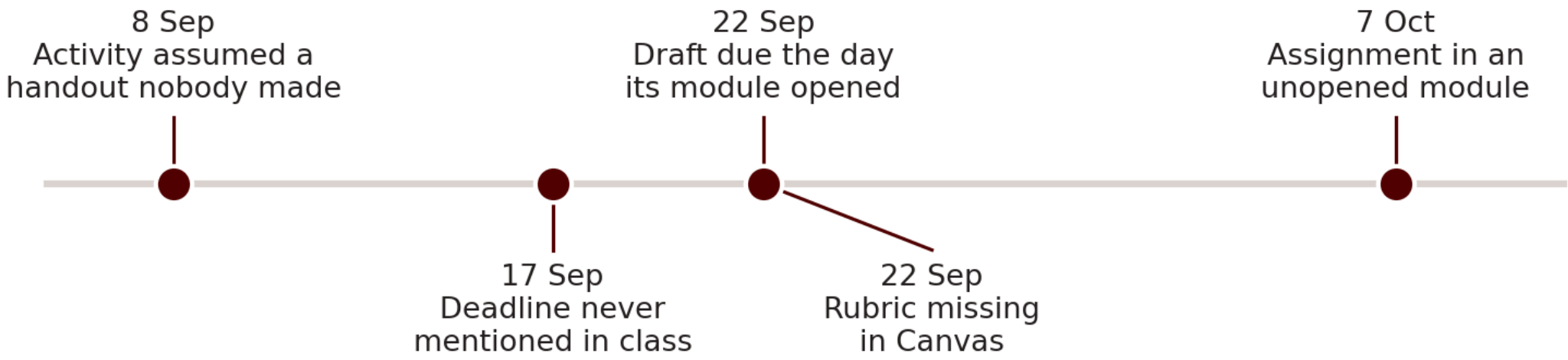
Every activity has a Bloom's Level

## Checks

Every rule is a program that can fail

# Problems that traceability caught in GEOG 476 this semester

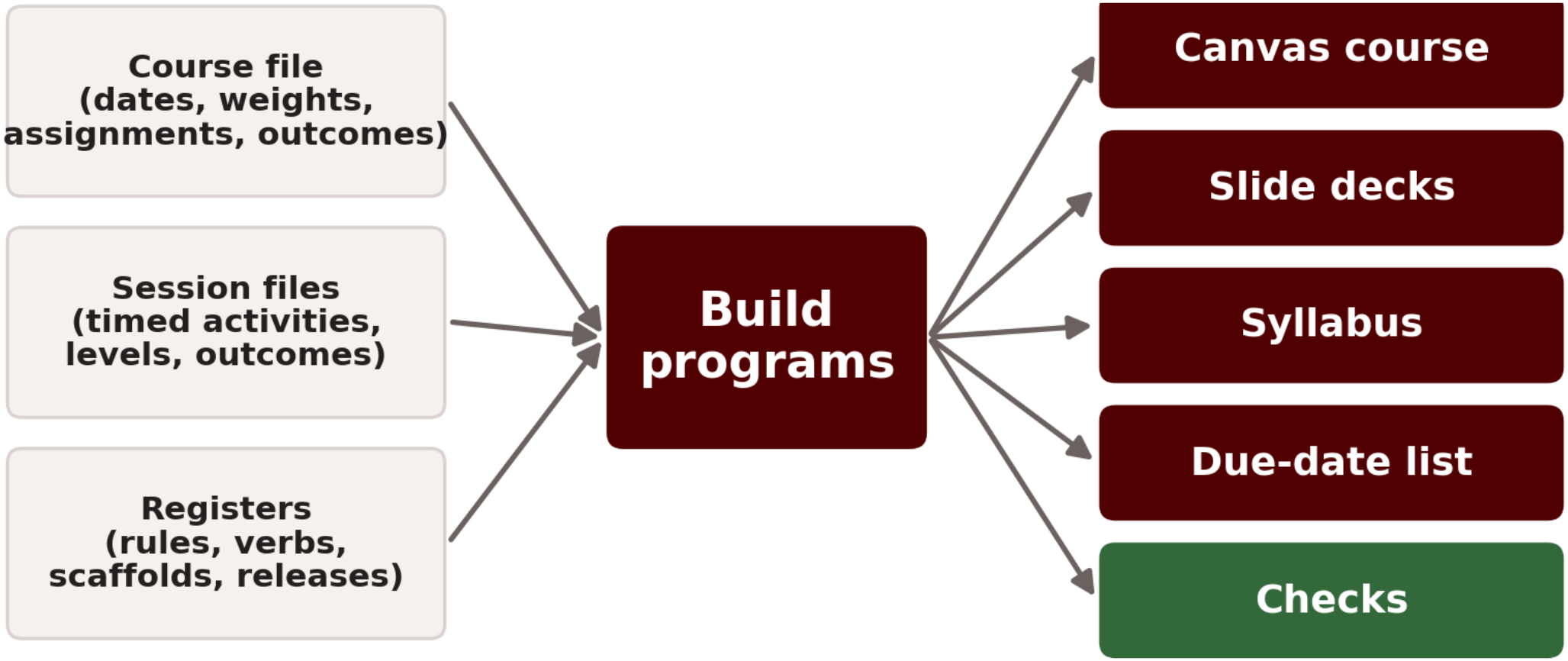
Each one looked correct on its own page.



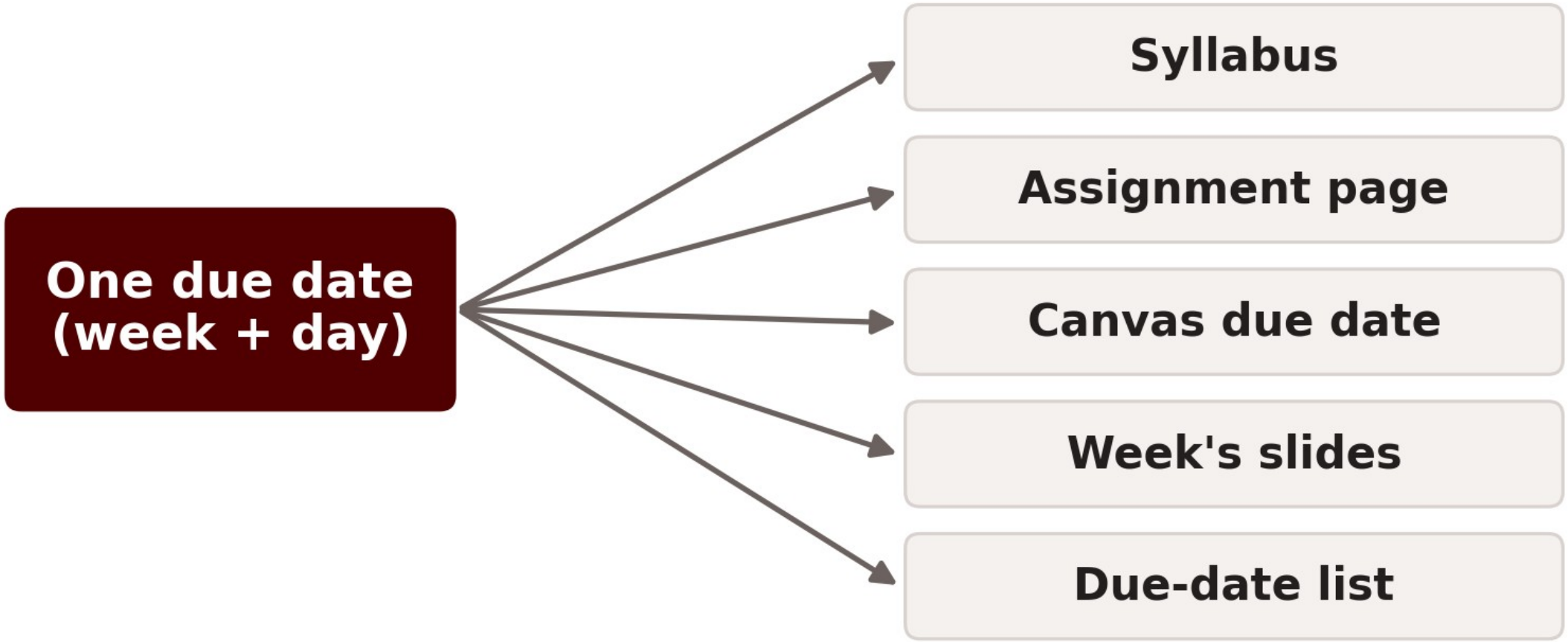
PART ONE

# The course is stored as data, not as pages

# How the GEOG 476 course files become what students see



# One due date, stored once and printed in five places

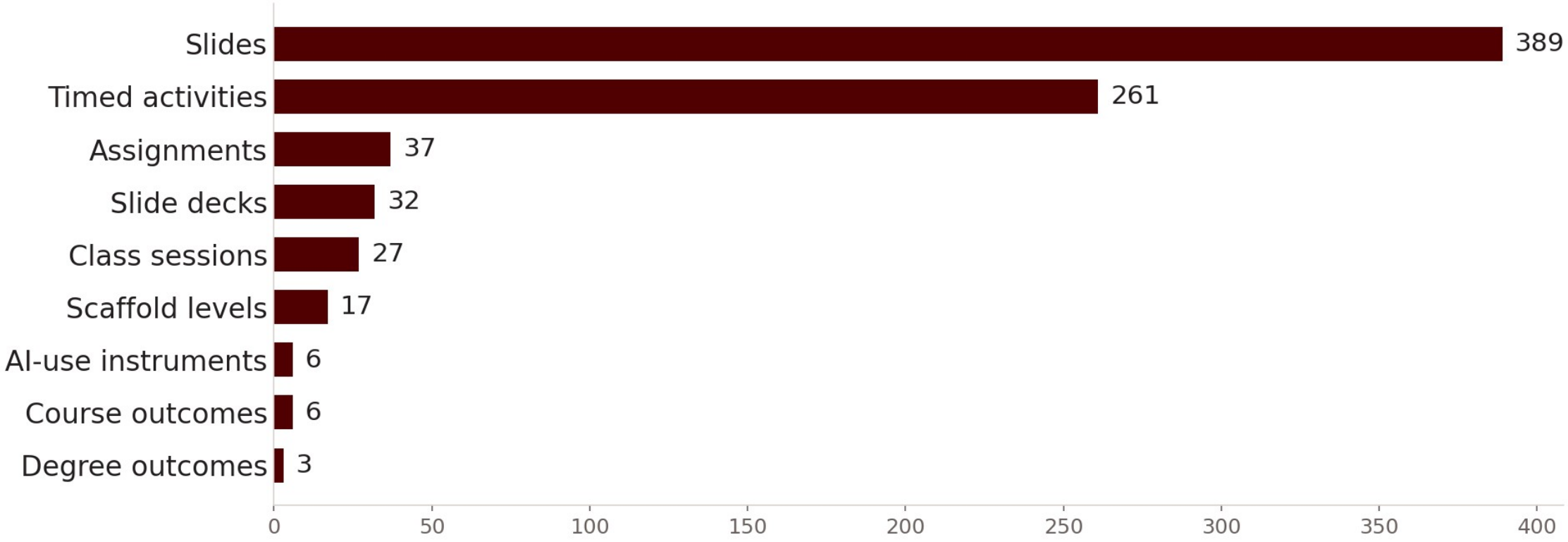


PART TWO

# Everything in the course is part of one graph

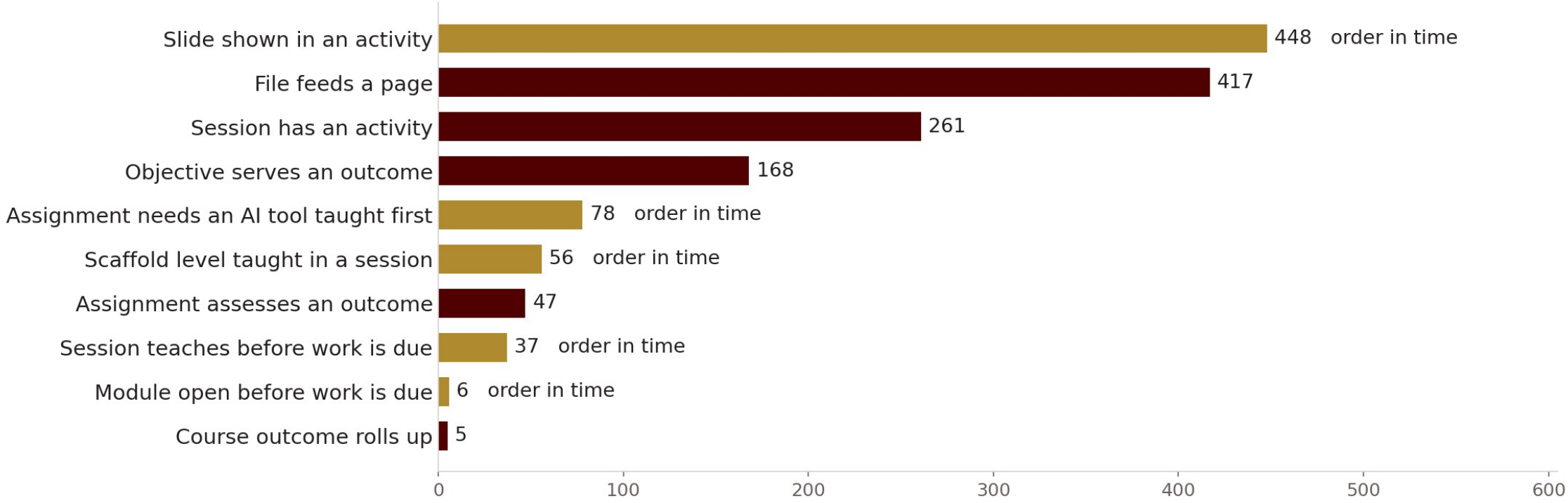
# What is in the GEOG 476 course graph

1,245 items, connected by 1,938 links.



# The kinds of links in the GEOG 476 course graph

Gold links say one thing must happen before another.

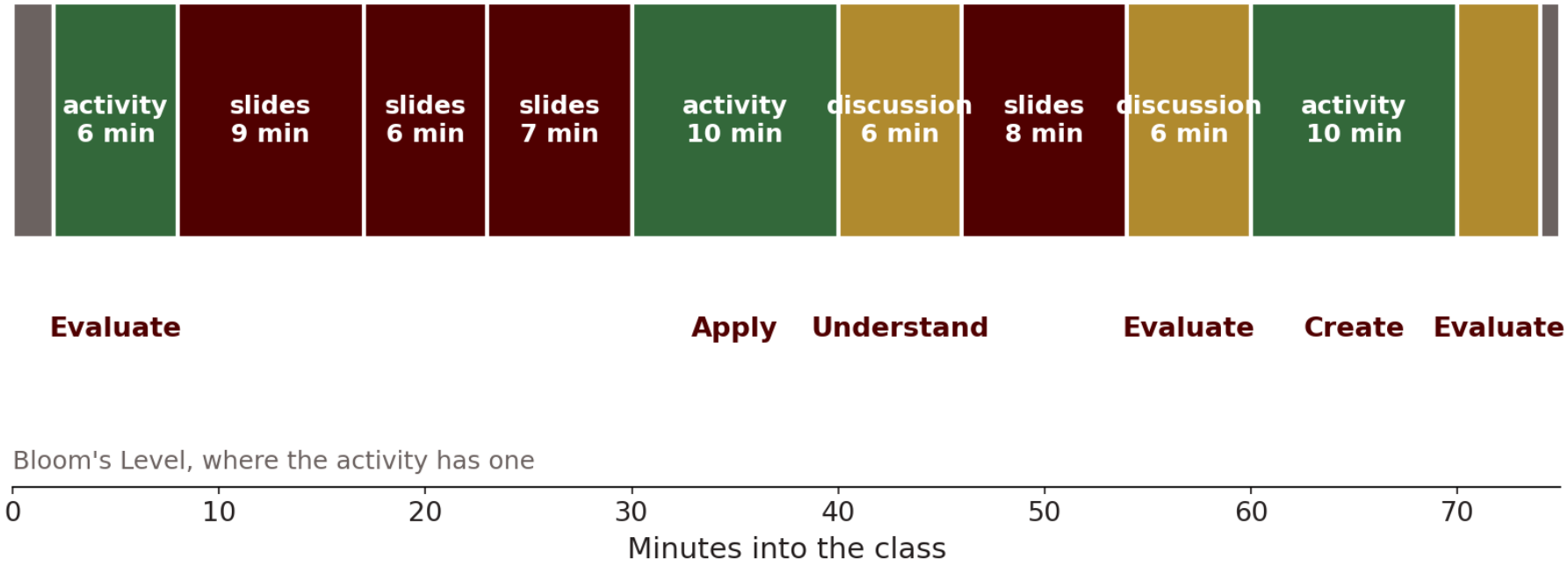


PART THREE

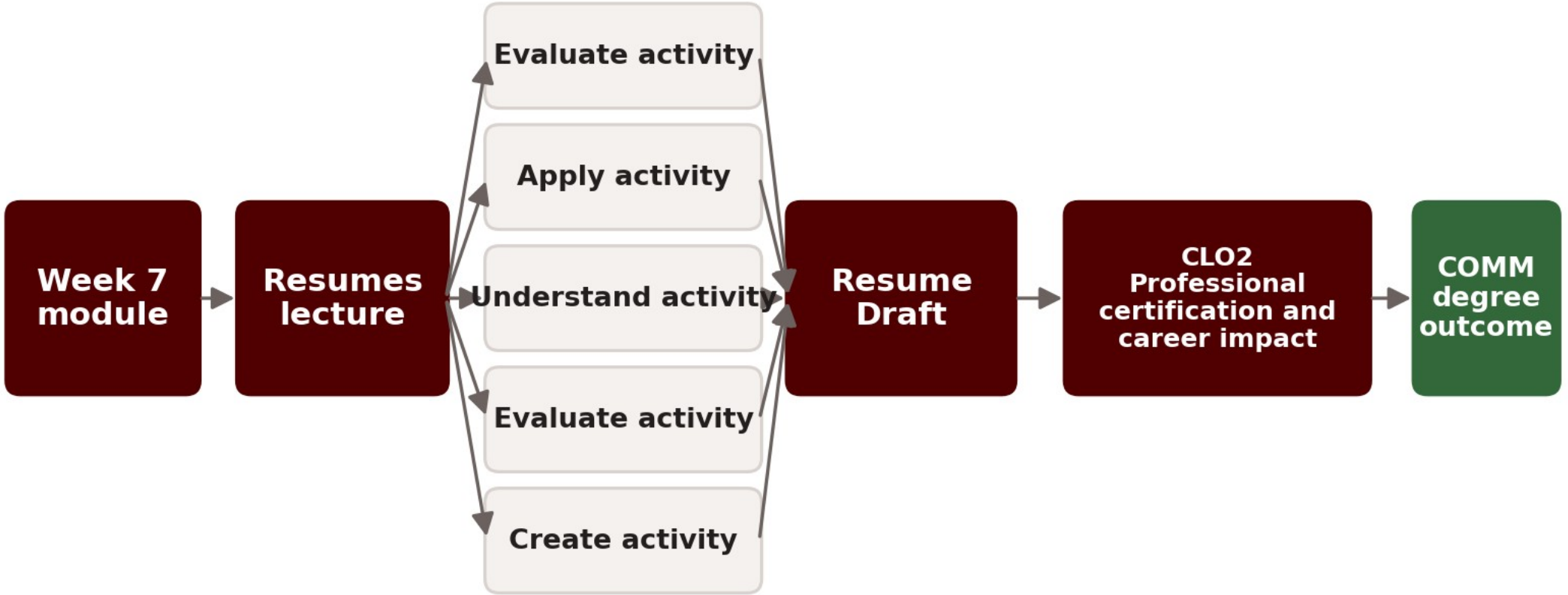
# Every activity has a Bloom's Level

# The Resumes lecture, as its session file describes it

Each block is a timed activity. The word below it is its Bloom's Level.

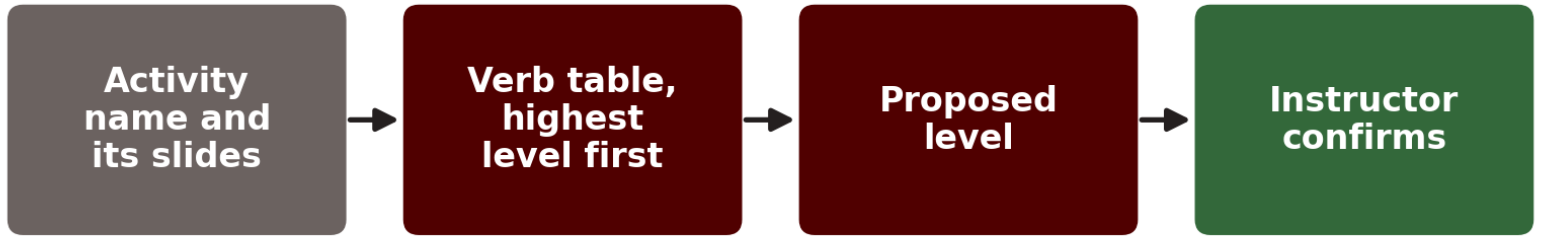


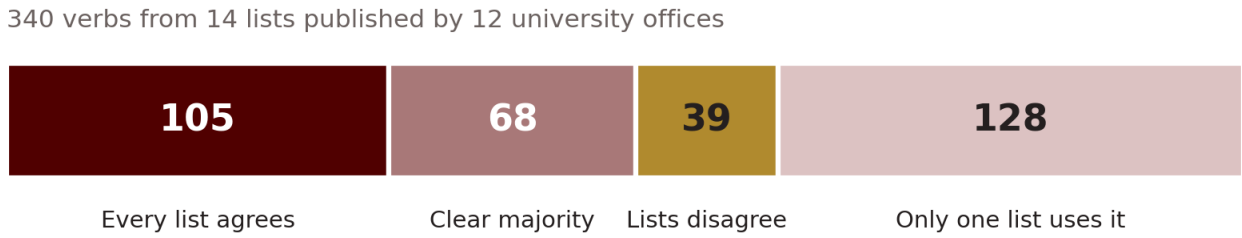
# The Resume Draft, followed from its module to a degree outcome



# Assigning a Bloom's Level, and where the verbs come from

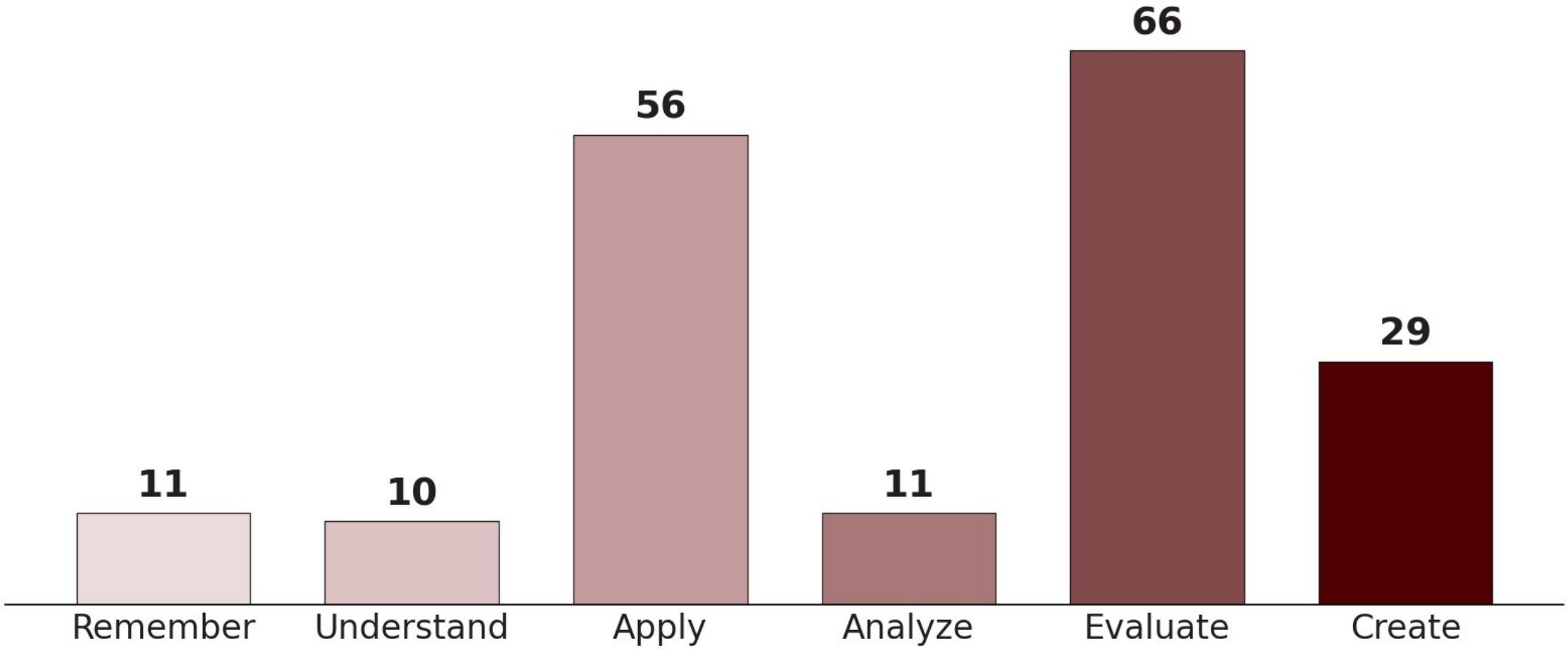
The Texas A&M list settles 12 of the 39 verbs the published lists disagree on.



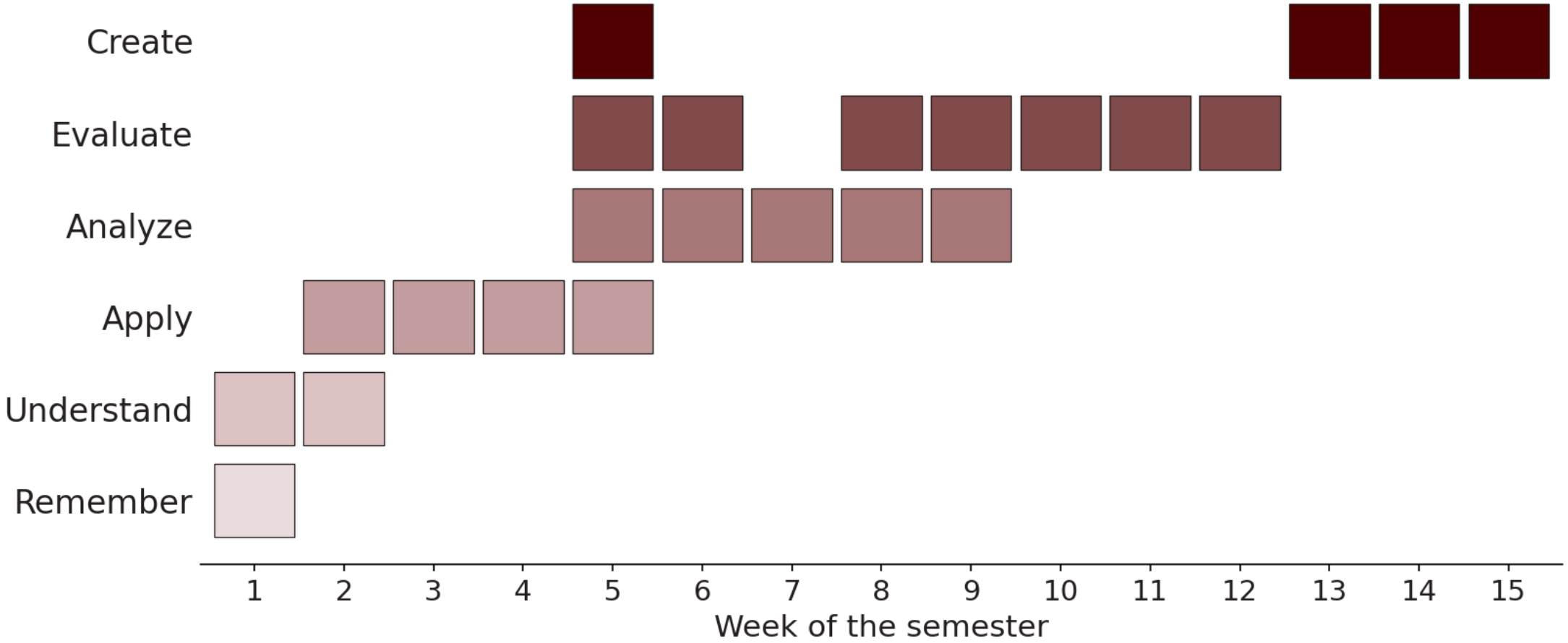
# Bloom's Levels of all 183 activities and assignments

2 levels are confirmed by me. 181 are still proposals.



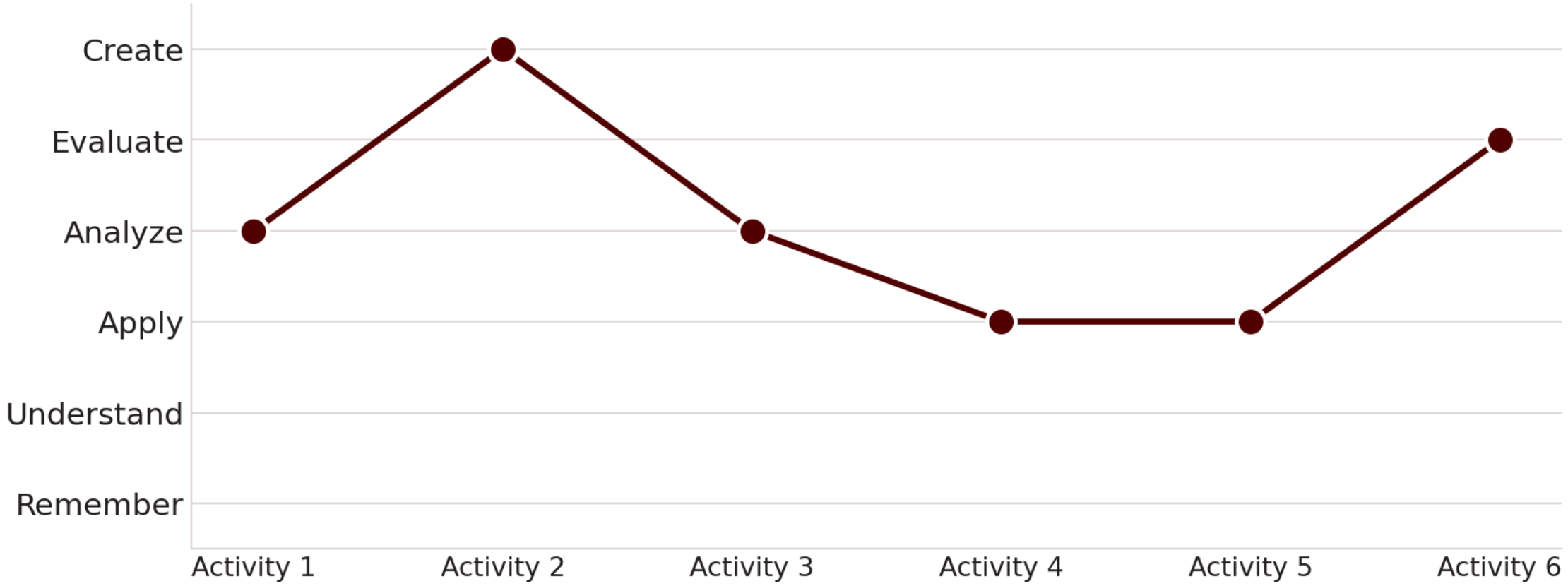
# The AI-use scaffold climbs from Remember to Create

A check refuses an AI activity at a level outside its weeks.



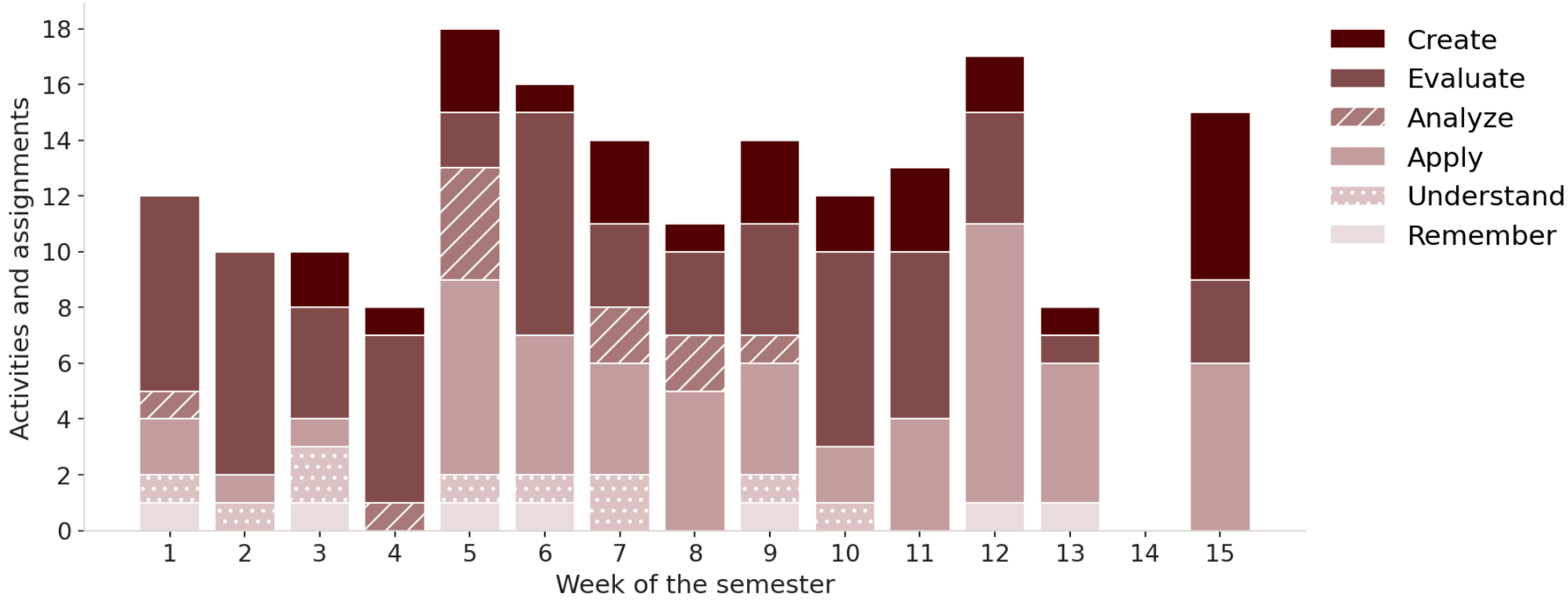
# Each class session has to climb in Bloom's Level

The Pilot Analysis studio, week 5. A session that stays at one level is refused.



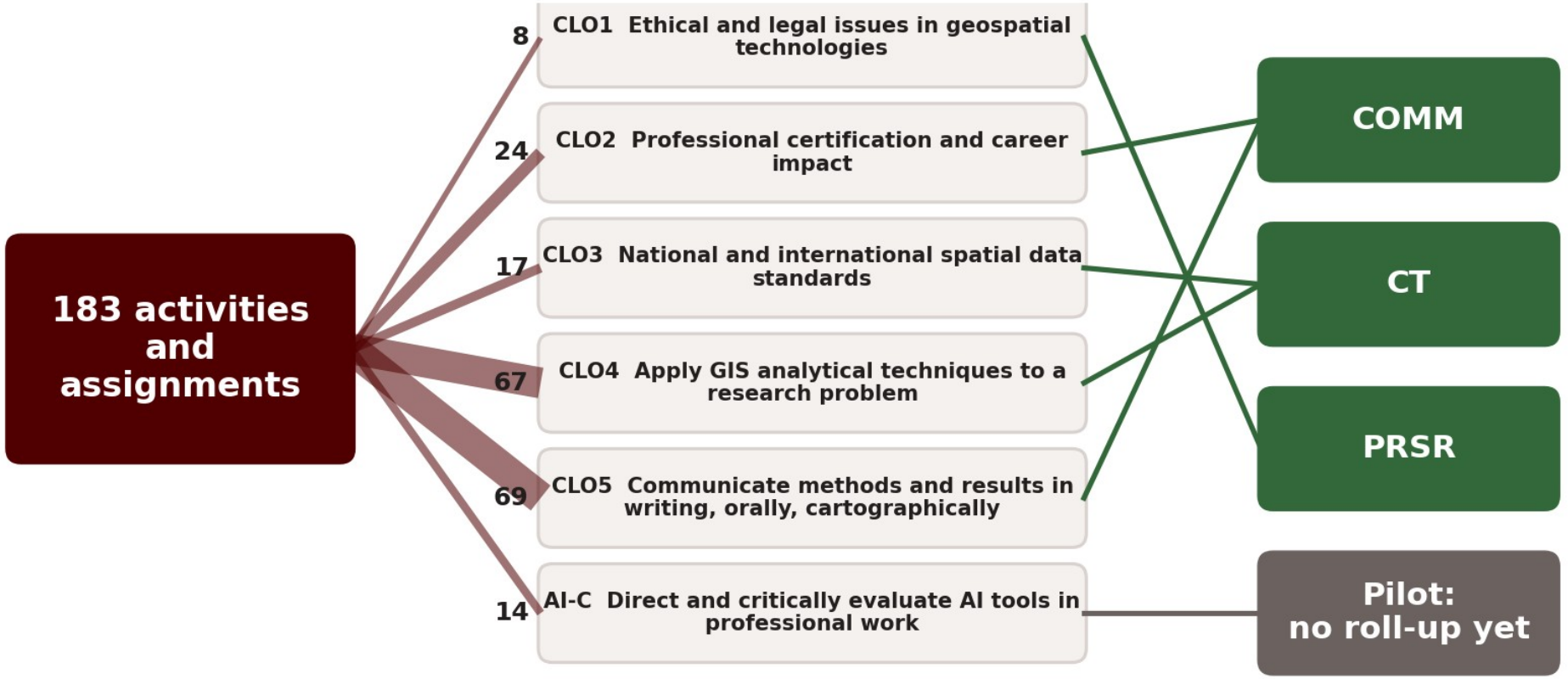
# Bloom's Levels of every activity and assignment by week

The whole course does not climb as neatly as the AI-use strand.



# How GEOG 476 activities reach degree outcomes

Line width is the number of activities and assignments on that path.

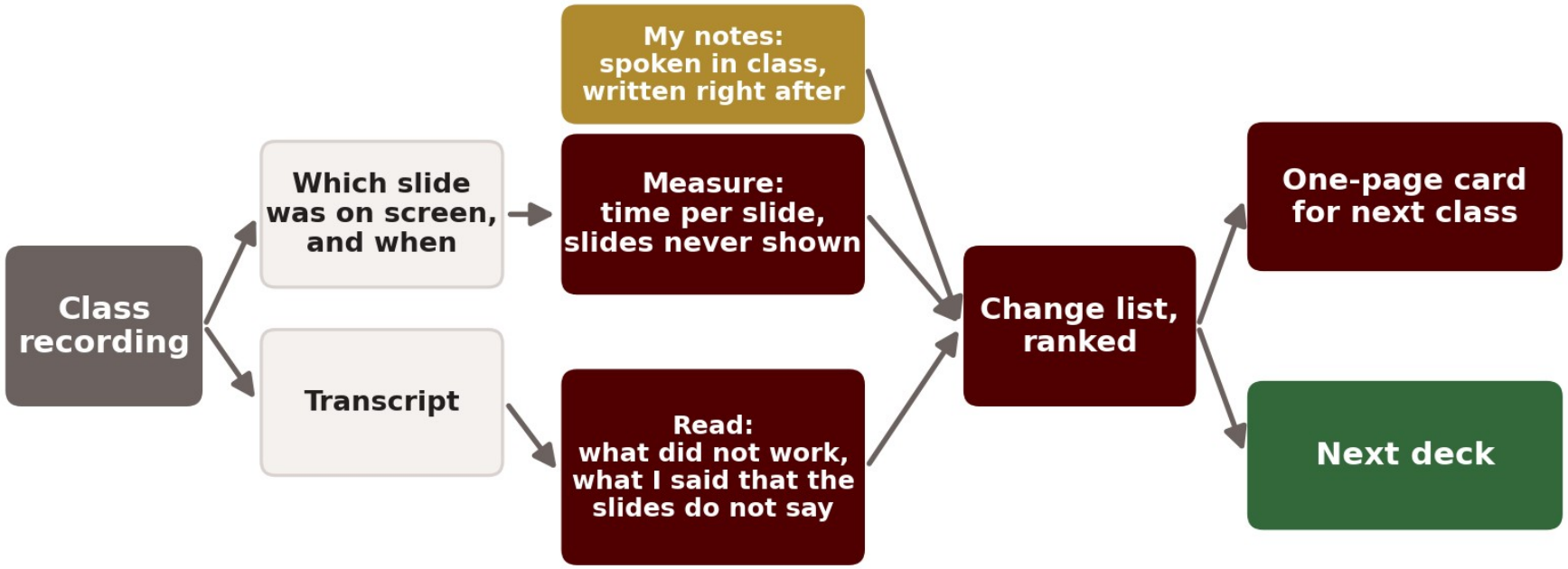


PART FOUR

# Every class recording is reviewed against the plan

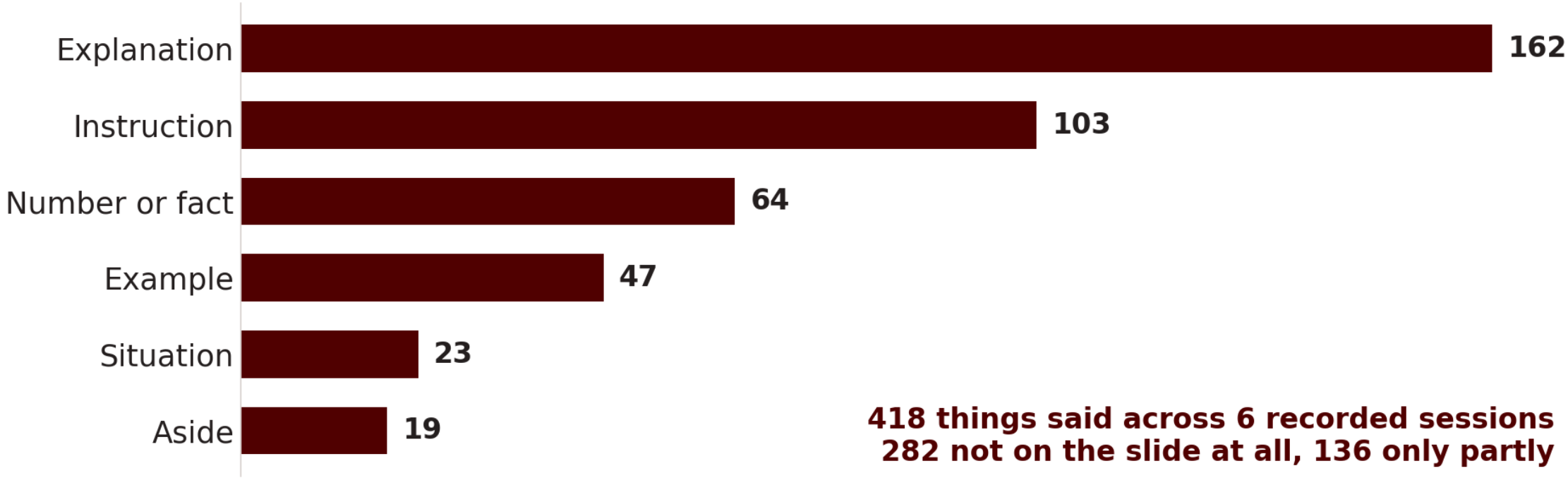
# How a GEOG 476 class recording becomes changes to the next class

The recording shows what the plan and the slides cannot.



# What I said in class that the slides did not carry

Each remark is logged with its time, the slide on screen, and where it should go.




**418 things said across 6 recorded sessions**
**282 not on the slide at all, 136 only partly**

# Off-the-cuff examples captured from the recordings

Good examples given once in class are written into the next version of the slides.

**Week 5 lecture, 31:40**

Use pictures instead of words: one picture each for infill, edge expansion and leapfrog growth.

**Week 5 lecture, 39:51**

Have AI run the same analysis with three different Python libraries. The same answer three ways is validation.

**Week 5 lecture, 20:23**

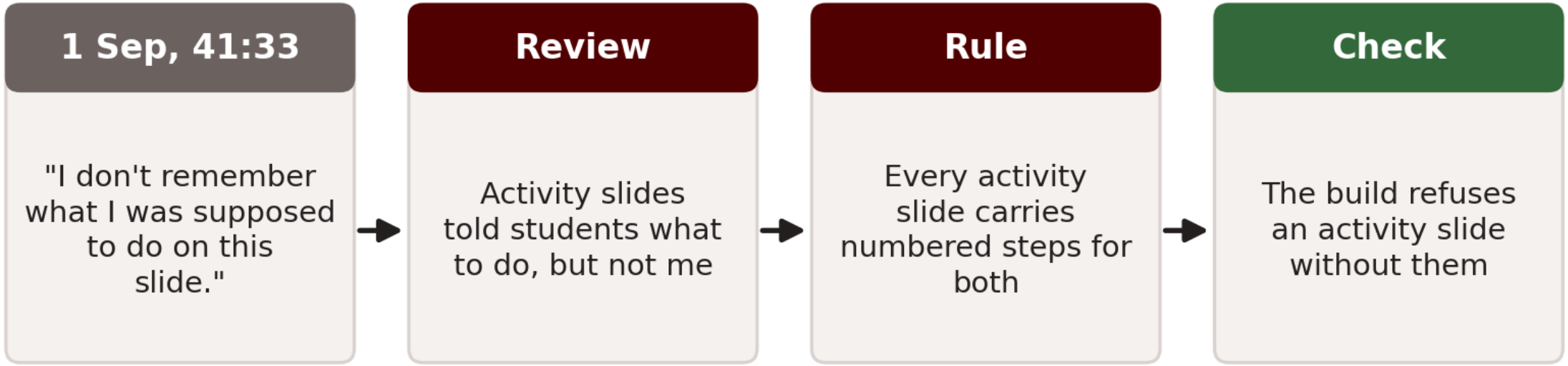
A narrowly defined component is good because you will know when you are done.

# Notes I take while teaching, by saying them aloud

The transcript keeps them. I add written notes in the hour after class.

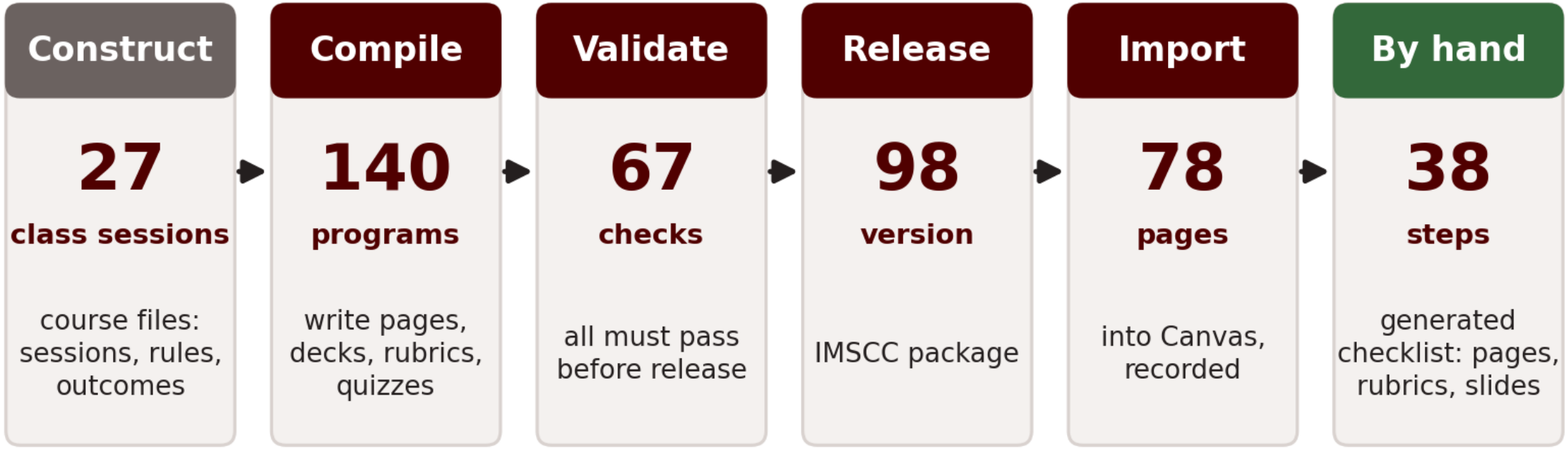
| Week 2 lecture, 1:08:43 | Week 3 studio | Week 5 studio |
| --- | --- | --- |
| "I need to make a note in the class of where the GitHub repository is put in public." | "I'll make a note in the slides here to have a link to some of those writing style resources." | "I should include that as a check in my class." |

# One moment in a recording that became a check



# How GEOG 476 is built, checked and released to Canvas

Nothing reaches Canvas until every check passes, and I still open it by hand.



QUESTIONS

# Thank you

Daniel W. Goldberg · daniel.goldberg@tamu.edu
These materials were generated by a human with the assistance of AI.